Supplemental Material: Mathematical Verification of the τ Framework Across Scales

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We verify the mathematical consistency of the temporal asymmetry parameter $\tau = 1 - F$ across four layers: (1) quantum channels, where the JRSWW bound is a proven theorem; (2) gravitational fields, where three independent routes motivate a conjecture; (3) classical systems, where the bound reduces to Bayesian retrodiction; and (4) stochastic thermodynamics, where the Crooks fluctuation theorem provides the bridge. Each layer is assigned an honest status label.

LAYER 1: QUANTUM CHANNELS

Status: [THEOREM] — mathematically proven.

The JRSWW bound

For a quantum channel $\mathcal{N}$ (CPTP map) and states ρ, σ with $\rho \ll \sigma$, the quantum relative entropy (QRE) satisfies the data processing inequality (DPI):

$$D(\rho\|\sigma) \geq D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)). \tag{1}$$

Define the QRE drop:

$$\Sigma \;\equiv\; D(\rho\|\sigma) - D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)) \;\geq\; 0. \tag{2}$$

Junge, Renner, Sutter, Wilde, and Winter [1] proved the strengthened DPI: there exists a rotated Petz recovery map $\tilde{\mathcal{R}}$ such that

$$F\big(\rho,\, \tilde{\mathcal{R}} \circ \mathcal{N}(\rho)\big) \;\geq\; \exp(-\Sigma/2), \tag{3}$$

where $F(\rho, \omega) = \left\|\sqrt{\rho}\sqrt{\omega}\right\|_1$ is the Uhlmann fidelity [2]. An earlier version with F^2 on the left was obtained by Fawzi and Renner [3].

The temporal asymmetry parameter is defined as:

$$\tau \;\equiv\; 1 - F(\rho,\, \mathcal{R} \circ \mathcal{N}(\rho)), \tag{4}$$

so the JRSWW bound becomes the τ bound:

$$\tau \;\leq\; 1 - \exp(-\Sigma/2). \tag{5}$$

The equivalence chain

The following conditions are equivalent for a channel $\mathcal{N}$ with reference state σ [3–5]:

$$\tau = 0 \iff \Sigma = 0 \iff I(A;E|B)_\omega = 0 \iff \text{QMC}, \tag{6}$$

where ω_{ABE} is the purification of the channel output, $I(A;E|B)$ is the quantum conditional mutual information (QCMI), and QMC denotes the quantum Markov chain condition $\omega_{ABE} = \mathcal{R}_{B\to BE}(\omega_{AB})$.

The proof assembles three known results:

- $\Sigma = 0 \Leftrightarrow$ exact Petz recovery (Petz [4]).
- $\Sigma = 0 \Leftrightarrow I(A;E|B) = 0$ (Hayden–Jozsa–Petz–Winter [5]).
- $I(A;E|B) = 0 \Leftrightarrow$ approximate recovery with fidelity controlled by QCMI (Fawzi–Renner [3]).

Numerical verification

We numerically tested the τ bound (5) on three standard quantum channels: amplitude damping ($\gamma \in [0.01, 0.99]$, step 0.01), dephasing ($p \in [0.01, 0.99]$), and depolarizing ($p \in [0.01, 0.99]$). For each of 99 noise parameters, 20 random qubit states were generated (Haar-uniform), yielding $3 \times 99 \times 20 = 5{,}940$ test cases.

Result.—Zero violations of Eq. (5) were found. In all 5,940 cases, $\tau_{\text{measured}} \leq 1 - \exp(-\Sigma/2)$.

Dephasing anomaly.—For the dephasing channel with maximally mixed reference state $\sigma = I/2$, the identity map achieves strictly higher fidelity than the standard Petz map. This is because the Petz map with $\sigma = I/2$ reduces to $\mathcal{R} = \mathcal{N}^\dagger \circ (\cdot) = \mathcal{N} \circ (\cdot)$ (the dephasing channel is self-adjoint), so applying the Petz map means applying the dephasing channel a second time. The identity map, which does nothing, trivially outperforms double-dephasing. This does not violate the JRSWW bound, which guarantees only that the *rotated* Petz map is near-optimal.

LAYER 2: GRAVITATIONAL CHANNELS

Status: [MOTIVATED CONJECTURE] — well-supported but not proven.

The identification

Paper 2 of this series [6] proposes the gravitational entropy production:

$$\Sigma_{\text{grav}} = -\ln(-g_{00}) = \frac{r_s}{r} \quad \text{(exponential metric)}, \tag{7}$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius. Three independent derivations converge on this identification.

Route 1: Modular flow

Dorau and Much [7] showed that modular flow in algebraic QFT derives the Einstein field equations from quantum relative entropy. Their construction defines a local QRE density whose integral over a causal diamond gives the total entropy production. For a static spherically symmetric spacetime, the fractional QRE loss between the diamond at infinity and one at radius r evaluates to:

$$\frac{D_{\text{in}} - D_{\text{out}}}{D_{\text{in}}} = \frac{r_s}{r}. \tag{8}$$

This is exact, not an approximation.

Route 2: Gravitational Landauer principle

Herrera [8] derived the minimum energy cost of erasing one bit of information at radius r in a gravitational field, accounting for the Tolman redshift factor:

$$W_{\text{erase}}(r) = k_{\text{B}} T_{\text{local}} \ln 2 = \frac{k_{\text{B}} T_\infty}{\sqrt{-g_{00}}} \ln 2. \tag{9}$$

The entropy production per erasure, measured from infinity, is:

$$\Sigma_{\text{Landauer}} = -\ln(-g_{00}), \tag{10}$$

matching Eq. (7).

Route 3: Bosonic loss channel

Ahmadi and Fuentes [9] showed that a bosonic field mode propagating in a static spacetime undergoes a pure-loss bosonic channel with transmissivity $\eta = -g_{00}$. For a pure-loss channel, the QRE drop between coherent state inputs is [10]:

$$\Sigma_{\text{bosonic}} = -\ln \eta = -\ln(-g_{00}), \tag{11}$$

again matching Eq. (7).

Critical caveats

No first-principles CPTP map.—The JRSWW bound (3) requires a CPTP map $\mathcal{N}$ satisfying $\Delta D = \Sigma$. No single, first-principles CPTP map has been proven to give $\Delta D = -\ln(-g_{00})$ exactly. Each route above extracts Σ_{grav} from a different physical construction; none provides a universal gravitational channel.

Dimensional obstruction.—Finite-dimensional (qubit) channels satisfy $\Delta D \leq \ln d$, where d is the Hilbert space dimension. Since $\Sigma_{\text{grav}} = r_s/r$ is unbounded as $r \to r_s$, the gravitational channel must be infinite-dimensional (bosonic). This is consistent with Route 3 but means qubit numerics cannot probe the strong-field regime.

Saturation is impossible for $\Sigma > 0$.—The Golden-Thompson inequality [11] $\text{Tr}(e^A e^B) \geq \text{Tr}(e^{A+B})$ makes the JRSWW bound strictly non-saturating whenever $[\rho, \sigma] \neq 0$ and $\Sigma > 0$. The exponential metric $g_{00} = -e^{-r_s/r}$ cannot be derived from bound saturation alone.

What survives.—The identification

$$\frac{c_{\text{eff}}}{c} = \exp(-\Sigma_{\text{grav}}/2) = F_{\text{bound}} \tag{12}$$

is a mathematical identity following from the definition $\Sigma_{\text{grav}} = 2|\Phi|/c^2$ and the definition of the effective speed of light. It does not depend on saturation and holds regardless of the channel details.

Alternative route via quantum hypothesis testing.—Hayashi and Yamasaki [12] define Σ directly from quantum states via $\Sigma_{\text{grav}} = D(\rho(z_1) \| \rho(z_2))$, where $\rho(z)$ is the field state at spacetime point z. This bypasses the need for a CPTP map entirely, at the cost of requiring well-defined local states.

LAYER 3: CLASSICAL LIMIT

Status: [THEOREM] — proven as special case of Layer 1.

Statement

Theorem 1 (Classical τ Bound)**.** *For any classical channel (stochastic matrix) P with prior distribution p and posterior $q = Pp$, the temporal asymmetry parameter satisfies:*

$$\tau_{cl} \;\leq\; 1 - \exp\!\big(-\Delta D_{\text{KL}}/2\big), \tag{13}$$

where $\Delta D_{\text{KL}} = D_{\text{KL}}(p\|s) - D_{\text{KL}}(Pp\|Ps)$ is the KL divergence drop, s is the reference distribution, and τ_{cl} is computed using the Bayesian retrodiction map.

Proof: reduction from the quantum case

The proof consists of showing that each quantum object in the JRSWW bound reduces to its classical counterpart when all states commute.

Step 1: Petz map → Bayes' rule.—For commuting states (diagonal density matrices), the Petz recovery map [4, 13]

$$\mathcal{R}_{\sigma,\mathcal{N}}(\omega) = \sigma^{1/2}\, \mathcal{N}^\dagger\!\big(\mathcal{N}(\sigma)^{-1/2}\, \omega\, \mathcal{N}(\sigma)^{-1/2}\big)\, \sigma^{1/2} \tag{14}$$

reduces to the Bayesian inverse. Explicitly, with $\sigma = \mathrm{diag}(s_1, \ldots, s_n)$ and stochastic matrix $P_{j|i}$:

$$\mathcal{R}(P, s)_{i|j} = \frac{s_i\, P_{j|i}}{(Ps)_j} = P(i|j), \tag{15}$$

which is Bayes' rule.

Step 2: QRE → KL divergence.—The quantum relative entropy $D(\rho\|\sigma) = \mathrm{Tr}[\rho(\ln\rho - \ln\sigma)]$ [14] reduces to the Kullback–Leibler divergence $D_{\mathrm{KL}}(p\|s) = \sum_i p_i \ln(p_i/s_i)$ when $\rho = \mathrm{diag}(p)$ and $\sigma = \mathrm{diag}(s)$.

Step 3: Uhlmann fidelity → Bhattacharyya coefficient.—For probability distributions p and q, the Uhlmann fidelity becomes [2]:

$$F(p, q) = \sum_i \sqrt{p_i\, q_i} \equiv \mathrm{BC}(p, q), \tag{16}$$

the Bhattacharyya coefficient.

Step 4: Assembling.—Substituting Steps 1–3 into the JRSWW bound (3) yields Eq. (13) directly. □

Consequences

The classical τ bound applies to any system describable by a stochastic matrix:

- Thermodynamic processes (heat engines, molecular motors).
- Neural dynamics (transition matrices between brain states).
- Chemical reaction networks (master equations).

No additional assumptions beyond the existence of a classical channel and a reference distribution are needed.

LAYER 4: STOCHASTIC THERMODYNAMICS BRIDGE

Status: [BRIDGE EXISTS] — mathematically sound via the Crooks fluctuation theorem.

The problem

Stochastic thermodynamics uses the Schnakenberg entropy production [15]

$$\Sigma_{\mathrm{Sch}} = \sum_{i \neq j} J_{ij}\, \ln \frac{W_{ij}\, p_i}{W_{ji}\, p_j}, \tag{17}$$

where W_{ij} are transition rates and J_{ij} are probability currents. This is *not* the same as the JRSWW quantity Σ_{DPI} from Eq. (2). The Schnakenberg entropy production counts dissipation along edges of a graph; the JRSWW quantity measures global KL divergence drop under a channel. Direct identification would be incorrect.

Solution: Crooks on path space

The bridge is provided by the Crooks fluctuation theorem [16] applied to trajectory space. For a system driven by protocol $\lambda(t)$, each individual trajectory γ carries a stochastic entropy production:

$$\Sigma_{\mathrm{traj}}(\gamma) = \ln \frac{P_{\mathrm{fwd}}(\gamma)}{P_{\mathrm{rev}}(\bar{\gamma})}, \tag{18}$$

where $P_{\mathrm{fwd}}(\gamma)$ is the probability of trajectory γ under the forward protocol and $P_{\mathrm{rev}}(\bar{\gamma})$ is the probability of the time-reversed trajectory under the reversed protocol.

The Crooks theorem states [16]:

$$\left\langle e^{-\Sigma_{\mathrm{traj}}} \right\rangle = 1. \tag{19}$$

Now treat the forward and reverse path ensembles as probability distributions P_{fwd} and P_{rev} on trajectory space. The Bhattacharyya coefficient between them satisfies (by Jensen's inequality applied to Eq. (19)):

$$\mathrm{BC}(P_{\mathrm{fwd}}, P_{\mathrm{rev}}) \;\geq\; \exp\!\big(-\Sigma_{\mathrm{Sch}}/2\big), \tag{20}$$

where $\Sigma_{\mathrm{Sch}} = \langle \Sigma_{\mathrm{traj}} \rangle$ is the mean entropy production, equal to the KL divergence $D_{\mathrm{KL}}(P_{\mathrm{fwd}}\|P_{\mathrm{rev}})$ on path space.

Since the Bhattacharyya coefficient is the classical fidelity (Layer 3, Eq. (16)), the classical τ bound applies:

$$\tau_{\mathrm{path}} \;\leq\; 1 - \exp\!\big(-\Sigma_{\mathrm{Sch}}/2\big). \tag{21}$$

Esposito–Van den Broeck decomposition

The total entropy production decomposes as [17]:

$$\Sigma_{\mathrm{total}} = \Sigma_{\mathrm{ad}} + \Sigma_{\mathrm{na}}, \tag{22}$$

where the adiabatic contribution

$$\Sigma_{\mathrm{ad}} = D_{\mathrm{KL}}\big(p(t) \,\big\|\, p_{\mathrm{eq}}(\lambda(t))\big) \tag{23}$$

measures the distance between the instantaneous distribution $p(t)$ and the equilibrium distribution $p_{\mathrm{eq}}(\lambda(t))$ for the current protocol value. This is precisely a KL divergence and is therefore the JRSWW-compatible piece: it equals the QRE drop for the thermalization channel. The nonadiabatic contribution Σ_{na} captures the lag between the system's response and the driving protocol.

Application: brain entropy production

Sanz Perl et al. [18] measured entropy production in human brain dynamics using the KL divergence between forward and time-reversed neural trajectories:

$$\Sigma_{\mathrm{brain}} = D_{\mathrm{KL}}(P_{\mathrm{fwd}}\|P_{\mathrm{rev}}). \tag{24}$$

They found Σ_{brain} monotonically correlated with consciousness level: highest in wakefulness, reduced under anesthesia, lowest in unresponsive states. The τ bound (21) applies to these measurements, giving an upper limit on the retrodictability of brain states.

IIT's Φ is not Σ

It is tempting to identify integrated information Φ (from Integrated Information Theory [19]) with entropy production Σ. This identification is wrong. Explicit counterexamples exist in both directions:

- High Φ, low Σ: a system with strong integration but operating near equilibrium (e.g., a balanced neural circuit with slow dynamics).
- High Σ, low Φ: a highly dissipative but modular system (e.g., two independent heat engines running in parallel).

Φ measures the irreducibility of a system's causal structure; Σ measures thermodynamic irreversibility. These are distinct quantities that can vary independently.

SUMMARY

Table I summarizes the verification across all four layers.

The τ parameter has a secure mathematical foundation (Layer 1), a well-motivated gravitational extension (Layer 2), a rigorous classical limit (Layer 3), and a sound connection to stochastic thermodynamics (Layer 4).

What remains open:

1. Layer 2 requires either a first-principles CPTP map for gravity or a proof that the quantum hypothesis testing route reproduces $\Sigma_{\text{grav}} = -\ln(-g_{00})$.
2. Layer 4 would benefit from an explicit derivation of the nonadiabatic contribution's effect on the τ bound.

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TABLE I. Mathematical verification of the τ bound across four layers. Each layer is assigned a status label reflecting the current state of proof.

Layer	**Domain**	**Status**	**Key equation**	**What it proves**
1	Quantum channels	THEOREM	$F \geq \exp(-\Sigma/2)$	τ bound holds universally
2	Gravitational fields	MOTIVATED CONJECTURE	$\Sigma_{\text{grav}} = -\ln(-g_{00})$	3 routes converge
3	Classical systems	THEOREM	$\tau_{\text{cl}} \leq 1 - \exp(-\Delta D_{\text{KL}}/2)$	Special case of Layer 1
4	Stochastic thermo	BRIDGE EXISTS	$\tau_{\text{path}} \leq 1 - \exp(-\Sigma_{\text{Sch}}/2)$	Via Crooks FT